# DERMA ASSIST - DETAILED TECHNICAL IMPLEMENTATION PLAN

Academic Project Exhibition Edition | Multimodal Clinical Architecture

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Scope: Full-Stack Multimodal Clinical Triage AI Benchmark: HAM10000 / ISIC + DermNet + Gemini

## 1. Executive Summary & Core Paradigm

Derma Assist is an advanced clinical dermatology co-pilot designed for high-impact academic project exhibitions. Unlike generic 'black-box' skin classification apps that blindly output probability scores on isolated photos, Derma Assist implements an authentic clinical methodology: it combines a structured, branching patient intake questionnaire with deep convolutional neural networks trained on genuine clinical benchmarks (HAM10000 and ISIC archives), visual explainability (Grad-CAM), and a mathematical Bayesian multimodal fusion algorithm with medical safety overrides.

## 2. Real-World Machine Learning Pipeline (HAM10000 / ISIC)

- Gold-Standard Dataset: Uses the HAM10000 dataset (10,015 real clinical dermoscopic captures from Harvard Dataverse/ViDIR) covering 7 core diagnostic categories:
  1. Melanoma (mel) - Malignant skin cancer
  2. Melanocytic Nevus (nv) - Benign moles
  3. Basal Cell Carcinoma (bcc) - Common epidermal malignancy
  4. Actinic Keratoses (akiec) - Precancerous dysplastic lesions
  5. Benign Keratosis (bkl) - Seborrheic keratosis, solar lentigo
  6. Dermatofibroma (df) - Benign histiocytoma
  7. Vascular Lesions (vasc) - Hemangiomas, pyogenic granulomas
- PyTorch Training Pipeline: Standalone PyTorch script ('ml_pipeline/train_ham10000.py') implementing PyTorch Dataset/DataLoader, data augmentation (random horizontal/vertical flips, affine rotations, color jitter), AdamW optimizer with cosine annealing, and multi-class Cross-Entropy Loss with class weighting to resolve dataset imbalance.
- Live Web App Weights: Employs a pre-trained deep convolutional neural network checkpoint fine-tuned on dermoscopy for immediate, sub-50ms CPU/GPU inference with authentic feature representations.

## 3. Guided Multi-Section Branching Questionnaire Engine

To mirror genuine dermatological consultation, all mock demo buttons are replaced with an interactive clinical intake wizard:
- Section 1 (Chief Complaint & Location): Visual selection of primary presentation (Pigmented Spot/Mole, Red Itchy Rash, Scaly Plaque, Facial Acne/Bumps) and anatomical location (Face, Scalp, Trunk, Extremities, Palms/Soles).
- Section 2 (Timeline & Sensation): Onset duration (Acute <48h, Subacute 1-4w, Chronic >1m) and symptomatic profile (Pruritus, sharp pain, burning, tenderness, asymptomatic).
- Section 3 (Dynamic Branching): Dynamically reveals condition-specific inquiry:
  * Mole/Growth Branch: Comprehensive ABCDE assessment (Asymmetry, Border irregularity, Color variation, Diameter >6mm, Evolution, Bleeding).
  * Rash Branch: Morphology (annular/ring vs. diffuse hives) and contact triggers (new detergents, cosmetics, outdoor exposure, pets).
  * Scaly Plaque Branch: Silvery micaceous scale on extensor surfaces (psoriasis) vs. flexural folds (eczema).
  * Facial Branch: Flushing triggers, comedones, pustules.
- Section 4 (Emergency Red-Flag Screening): Immediate assessment of severe warning signs (high fever with blistering, rapid purple purpura, facial angioedema/breathing difficulty).

## 4. Bayesian Multimodal Fusion & Medical Safety Overrides

- Mathematical Fusion Algorithm: Computes posterior probability by multiplying prior image probabilities from the deep learning model by the clinical symptom likelihood multipliers derived from the questionnaire answers:
  $P(\text{Condition} \mid \text{Image}, \text{Symptoms}) \sim P(\text{Condition} \mid \text{Image}) * P(\text{Symptoms} \mid \text{Condition})$
- Clinical Safety Override: Medicine requires fail-safe guardrails. If a lesion appears visually borderline but the questionnaire reports spontaneous bleeding, rapid growth, or irregular dark border (ABCDE criteria), the system triggers an emergency clinical override, automatically elevating the risk to 'High Risk / Urgent Excisional Biopsy' and alerting the patient and dermatologist.

## 5. Visual Explainability via Grad-CAM

To satisfy evaluator scrutiny regarding black-box models, Derma Assist implements Gradient-Weighted Class Activation Mapping

(Grad-CAM). It projects spatial activation density into a Jet colormap (Blue = baseline tissue, Red/Yellow = peak attention), proving to judges that the model focuses on lesion perimeters, erythema centers, and pigment variegation rather than background skin artifacts or lighting.

## 6. Scalability, Extensibility & Future Development Roadmap

The architecture is decoupled into modular layers to facilitate effortless future expansion:
- Modular Backend Services: 'fusion_service.py', 'gemini_service.py', and 'report_service.py' communicate via standard Pydantic schemas, allowing new clinical algorithms (e.g. skin tone normalization, Fitzpatrick scale scoring) to be plugged in without refactoring.
- Dataset Extensibility: The PyTorch pipeline can ingest additional ISIC or DermNet archives by appending CSV rows to the dataset reader.
- REST API Architecture: FastAPI backend supports seamless deployment as a Docker container, mobile app backend (React Native / Flutter), or cloud microservice on AWS / Google Cloud Run.